

Name: _____

Class: _____

Date: _____

Lab 9: If This, Then That

Python Lab · Section 3.5–3.7 Booleans & Conditionals



Goal

Write programs that make decisions with if, elif, else, and Boolean operators.

Warm-up: Predict

```
x = 7
if x > 5 and x < 10:
    print("A")
elif x > 5:
    print("B")
else:
    print("C")
```

Predict the output. What would print for x = 12? x = 2?

AP Connection

- Python and/or/not = AP AND/OR/NOT.
- Python == compares; AP uses =.
- elif is a nested IF inside ELSE in AP pseudocode.

Build It

Step 1. Grade calculator: convert a score to a letter grade.

```
score = int(input("Score: "))
if score >= 90:
    print("A")
elif score >= 80:
    print("B")
# finish C and F
```

Step 2. Ticket price: \$8 for under 12 or 65+, otherwise \$12.

```
age = int(input("Age: "))
if age < 12 or age >= 65:
    price = 8
else:
    price = 12
print(price)
```

Step 3. Test with boundary values: 11, 12, 64, 65. Record the results.

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Build It (continued) · Level Up

Build It (continued)

Step 4. Choose-your-own adventure: write a short story with at least 3 decisions, including one nested if.

```
choice = input("Go left or right? ")
if choice == "left":
    choice2 = input("A door! Open it? ")
    if choice2 == "yes":
        print("...")
```

Level Up Challenges

Mild Even/odd checker using %.

Medium Leap year checker: divisible by 4 and not by 100, unless divisible by 400.

Spicy Rock-paper-scissors against a fixed computer choice, handling every case.

Python → AP Pseudocode

```
if score >= 90:
    grade = "A"
elif score >= 80:
    grade = "B"
else:
    grade = "F"
```

Write this as a nested IF/ELSE in AP pseudocode.

Exit Ticket

Why must the score ≥ 90 check come before score ≥ 80 ?
